

Credit Allocation Mix in Peer-to-Peer Lending: A Network Study on Bondora

Social Network Analysis - Group 7

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Table of Contents

1	Introduction	3
2	Dataset	5
2.1	Pre-Processing & Network Formation	6
2.2	Descriptive Statistics & Preliminary Analysis	6
3	Research Rationale	8
4	Results	9
4.1	Study 1 - QAP Linear Regression	9
4.2	Study 2 - ERGM	11
A	Supplements	12
A.1	Data Preprocessing Steps	12
A.2	Distributions of Variables across Samples	14
A.3	ERGM Results - Supplementary Tables	17
A.4	Source Code - Data Preprocessing	18
A.5	Source Code - QAP Linear Regression	24
A.6	Source Code - ERGM Network Analysis	28
	References	34
B	Technology Statement	36

List of Figures

1	Network Visualisation	7
2	Network Centrality Distribution Plots	8
3	QAP Coefficient Plots	9

4	ERGM Coefficient Plots	11
5	Number of Unique Users per Loan Type per Year	12
6	Distribution of Loan Amount across Samples	14
7	Distribution of Interest Rates across Samples	14
8	Distribution of Age across Samples	14
9	Distribution of Loan Duration across Samples	15
10	Distribution of Loan Purpose across Samples	15
11	Distribution of Credit Ratings across Samples	15
12	Distribution of Occupation Types across Samples	16

List of Tables

1	Hypotheses related to Study 2	5
2	Descriptive Statistics of Numeric Variables	6
3	Basic Network Descriptive Statistics	7
4		10
5	QAP Modelling Results	10
6		11
7	ERGM Modelling Results	11
8	ERGM Results in Probability Scale	17

1 Introduction

With the advent of online peer-to-peer (P2P) lending platforms, the traditional methods of financial intermediation have been usurped by individual choice; with no stringent third parties involved in transactions, individuals have greater power in sourcing credit. This presents new realities for individuals disenfranchised by the traditional financial system (Alistair Milne & Paul Parboteeah, 2017). Traditional financial literature shows that individually, investors exhibit behavioural biases (Agrawal, 2012; Ayal, Hochman, & Zakay, 2011) in how they choose their investments, however, numerous studies demonstrate that these behavioural biases are present among non-professional P2P borrowers. For example, Herzenstein, Dholakia, & Andrews (2011) and Lee & Lee (2012) suggest that users cluster around popular loans, exhibiting ‘herding’ behaviour. Literature focuses predominantly on funding success (Yao, Chen, Wei, Chen, & Yang, 2019), however, Ayal, Bar-Haim, & Ofir (2018) suggests that there is insufficient attention on how borrowers form their loan portfolios. As the authors indicate, this area of research is important in understanding why less professional investors deviate from known models of rational investor behaviour. Our study aims to model and understand how P2P borrowers’ behaviours influence what kinds of loans they acquire. In subsequent discussions, we consider investors as rational participants in credit markets, as described by theory, whereas borrowers refer to the users of P2P platforms.

Investigating behavioural biases, Ayal et al. (2018) find that investors’ familiarity with specific assets can make them seem less risky, making recognisable assets more attractive. However, such assets lead to undiversified portfolios. Evidence with P2P platforms supports the familiarity bias in lenders, as Galak, Small, & Stephen (2010) demonstrate that lenders choose borrowers who are similar across demographic attributes such as gender, occupation, and ethnicity. Moreover, Amar, Ariely, Ayal, Cryder, & Rick (2011) explains that borrowers show an aversion to holding a *large number of debts*, preferring to first reduce debt count, even if they had other debts with higher interest rates. This irrational behaviour demonstrates that individual borrowers face behavioural biases on the *type* and *number* of debts they select. Based on these notions, we formulate our first study, asking:

Research Question 1: *Do borrowers vary in their loan use type and frequency based on their individual and loan characteristics?*

This question is answered using linear Quadratic Assignment Procedure (QAP) regressions, as we can represent similarities across loan uses and attributes as matrices, weighted by loan use frequency, as inputs in the model.

Using data from LendingClub, Serrano-Cinca, Gutiérrez-Nieto, & López-Palacios (2015) study funding success, finding that the reported loan use is a significant factor explaining differences in default rates. Specifically, the authors argue that there is heterogeneity in the credit riskiness of individuals seeking specific loan purposes, indicating that credit rating can explain some variation in borrowers’ choice of loan use. Therefore, we formulate the following hypothesis:

Hypothesis 1: *Borrowers with the same credit ratings choose similar loan uses*

Supporting the familiarity bias, Ravina (2019) studies personal characteristics in P2P markets, finding that loans tend to be awarded to those sharing personal characteristics such as occupation. As Ayal et al. (2018) implies, this can be generalised to suggest that lenders’ loan grants reflect a wider network of similar borrower preferences, which can reflect the homogeneity in loan use

observed by Serrano-Cinca et al. (2015). Thus,

Hypothesis 2: Borrowers sharing the same occupation area tend to share loan uses

Further, behavioural finance suggests that individuals' financial decisions are not independent, but rather, shaped by bounded rationality and psychological preferences (Kahneman & Tversky, 1979). In the P2P context, Ayal et al. (2018) shows that these preferences can influence how borrowers choose loan types and how this can overlap with other borrowers' decisions. Using the Exponential Random Graph Model (ERGM) framework, we can identify how these behaviours manifest as structural regularities in a borrower-to-loan-use network. Therefore, we ask:

Research Question 2: *What behavioural mechanisms affect the structure of borrower-to-loan-use connections?*

The aforescribed section outlined that the familiarity bias can lead to similarities across borrower portfolios in terms of their characteristics. However, in P2P settings, this can manifest itself as familiarity *clustering*, wherein borrowers who have previously been granted certain loan types are more likely to repeat their choices, or select other categories that appear familiar to different borrowers. Herzenstein et al. (2011), studying borrower decisions on Prosper.com, identifies a 'strategic herding' behaviour, where individuals gravitate towards popular or socially-validated loans. Considering that loan popularity is directly observable through its number of bids, we assert that borrowers linked to one loan purpose are likely to connect with others sharing the same loan use. In network terms, this behaviour results in cyclical substructures where there are overlapping nodes of each of the bipartite partitions. We capture a conservative form of this cycle through the ERGM term `cycle(4)`, formulating:

Hypothesis 3: Borrowers who share one loan purpose are likely to share another

Further, Ayal & Zakay (2009), expands on the theory of Debt Account Aversion with the Theory of Perceived Diversification, explaining that borrowers' distress increases with not just by the number of debts, but also the perceived *distinctiveness* of each debt type, suggesting that in the P2P context, debt mentally differentiated by purpose would also be seen as distressing. Therefore, we assert that borrowers try to minimise loan distinctiveness by requesting loans of a single type. This can be captured by the `b1degree(1)` term, which models whether borrowers are only linked to one loan purpose. Therefore, we formulate:

Hypothesis 4: Borrowers exhibit preferences for minimal debt distinctiveness, selecting one loan use

Notwithstanding, credit choices relate to non-structural factors. Cooper, Gorbachev, & Luengo-Prado (2023) show that younger borrowers face constraints in the types of loans they can access in typical credit markets. The authors argue that this constraint shapes how these borrowers form loan portfolios, often concentrating around different use cases such as education, personal consumption, or small business loans. Considering the systemic differences in how age affects credit consumption, we assert that younger borrowers have different loan use portfolios. The ERGM term `b1cov("age")` can capture this by assessing how changes in age affect how borrowers form connections with different loan types. Therefore:

Hypothesis 5: Younger borrowers have different loan use mixes than older borrowers

Alongside age, the borrower's gender is known to be a significant factor in influencing credit use decisions. Aliano, Alnabulsi, Cestari, & Ragni (2023) study the role of gender and other factors on

borrowers’ probability of default on the Bondora.com, finding a persistent gender effect on different loan uses; women tend to default less on health, home, and business loans. Croson & Gneezy (2009) shows that these differences can be due to risk perception and self-selection effects, where women exhibit greater risk aversion, leading to different borrowing patterns. We expect that these differences in risk tolerance and perceived creditworthiness by lenders affect loan use composition in terms of gender. The term `b1nodematch("gender")` captures this effect by assessing homophily in loan use selection. Our premise is supported by a negative estimate for the term, indicating heterophily. Therefore:

Hypothesis 6: Gender differences lead to different loan use mixes

Table 1: Hypotheses related to Study 2

Hypothesis	Term	Explanation
H_3^a : Borrowers who share one loan purpose are likely to share another	<code>cycle(4)</code>	This captures the tendency of borrowers and loan uses to be cyclical, where people cluster around shared uses of loans.
H_4^a : Borrowers exhibit preferences for minimal debt distinctiveness, selecting one loan use	<code>b1degree(1)</code>	This captures the tendency of borrowers to prefer minimal distinctiveness in their loan uses, preferring to hold only one distinct type to minimise their perceived risk.
H_5^a : Younger borrowers have different loan use mixes than older borrowers	<code>b1cov("age")</code>	This captures the tendency for younger borrowers to choose different kinds of loans based on how differently they consume credit.
H_6^a : Gender differences lead to different loan use mixes	<code>b1nodematch("gender")</code>	This captures the idea that there are systemic differences in what kinds of loan uses are preferred the genders due to their perceived riskiness and creditworthiness.

Following this section, we outline the methodology, explaining our dataset and network construction. To facilitate this, descriptive statistics are analysed for both the dataset and resulting network. Subsequently, we elaborate on our choice of models, discussing how network models can be used to conduct the study. Then, modelling results are shown and explained to evaluate our hypotheses. Simultaneously, the robustness of the study is evaluated through the models’ diagnostics. Finally, we arrive at our conclusions, summarising the study, its results, and implications. Supporting items such as the source code and specific data processing steps are shown in section A.

2 Dataset

The study utilises publicly-available data from Bondora, a European P2P lending platform primarily operating in the Baltics and Spain. The dataset contains detailed information on defaulted

and non-defaulted loans granted to users between February 2009 and July 2021. Specifically, contained is a range of numeric, binary, categorical, and time-series attributes across 85,087 unique users and 179,235 individual loans. Since the company’s API is no longer accessible, we utilised a publicly available repository compiled by Manu Siddhartha (2021) on kaggle.com. The user made a series of API calls to collect the data. Following this, the dataset was processed according to Appendix A.1.

2.1 Pre-Processing & Network Formation

Initially, the dataset was filtered loans between 2014 and 2016 since the data is incomplete during other periods. Since we are studying interactions across two distinct set of nodes, a bipartite network was formed with unique users as the first partition and the purpose of loans as the second. This allows us to evaluate how borrowers interact with loan purposes; a one-mode projection onto users is not meaningful for P2P networks as it produces no discernible structure. For model efficiency and performance, we randomly sample 500 individuals from the larger sample. Ultimately, we have 500 and 9 nodes in the first and second partitions, respectively.

2.2 Descriptive Statistics & Preliminary Analysis

Table 2 summarises the numeric variables between the reduced and larger samples. First, borrowers take out relatively small loans with fairly high interest rates, possibly reflecting that the average borrower is deemed fairly risky. The average loan is quite long, at approximately 46 months out of a maximum 60. Figure 11 shows that the largest credit category is HR¹, which is the lowest possible rating. So, the typical borrower is relatively uncreditworthy. Moreover, the largest categories of loan use are “Other” and “Home Improvement”, suggesting these are largely personal. We observe minimal differences between the random sampling and the larger sample of individuals, per Table 2 and Figure 11, however, the study may be biased by the fact that the dataset contains only granted loans, rather than all bids for loans. This means our inferences about how borrowers behave is limited to successful bids. Further, the time period is relatively distant, meaning that recent structural changes to the P2P sector cannot be accounted for.

Table 2: Descriptive Statistics of Numeric Variables

DataFrame	Variable	Mean	Median	Std. Dev.	Min	Max
Initial Sample	Amount	2652.01	2125	2151.28	115	10630
Initial Sample	Interest	35.94	31	24.77	7.62	263.63
Initial Sample	Age	38.53	37	11.4	19	70
Initial Sample	LoanDuration	45.5	60	17.53	3	60
Reduced Sample	Amount	2637.64	2112.5	2112.2	170	10630
Reduced Sample	Interest	35.85	30	27.24	10.17	253.08
Reduced Sample	Age	38.73	37	11.34	21	69
Reduced Sample	LoanDuration	46.23	60	17.35	3	60

¹Bondora uses their own proprietary credit rating system where the best rating is AA and worst is HR (Bondora, 2024). The company bases these ratings on their calculated expected probability of loss on the loan. For example, someone rated AA ranges from a expected loss of 0% to 2%, whereas for HR this is 25% - >25%.

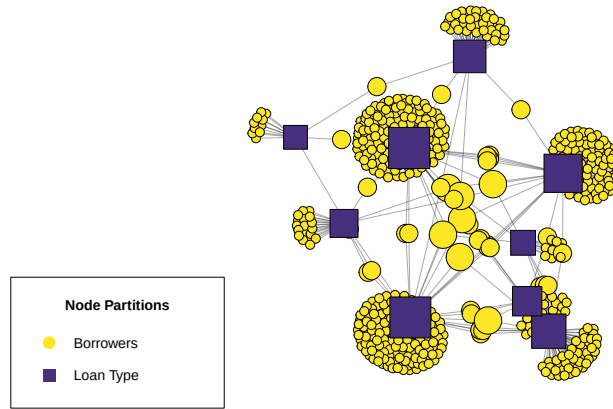
Table 3 shows a network density of 0.12, indicating that only 12% of all possible borrower–loan-type connections exist. This density suggests that borrowers typically engage with few loan types, reflecting specialization in borrowing behaviour. The centralization value of 0.47 further indicates an uneven distribution across loan types, where a few popular categories attract many borrowers while most remain peripheral. The mean distance of approximately three implies that borrowers are closely connected, typically separated by only three-to-four loan-use steps.

Centrality distributions reinforce this structure Figure 2. Betweenness is highest among loan-type nodes, though several borrowers also bridge distinct loan clusters. Closeness centrality remains low but scattered, suggesting localized clustering constrained by the bipartite structure. Degree distributions show wide variation among loan types, consistent with a hub-like structure dominated by a few popular uses. However, the second partition has relatively small degree dispersion. Finally, the homogeneous eccentricity distribution indicates that no borrowers are highly isolated, reflecting a compact and moderately cohesive network overall.

Table 3: Basic Network Descriptive Statistics

Statistic	Measure
Vertex Count	509.00
Edge Count	547.00
Density	0.12
Centralization	0.47
Mean Distance	3.66

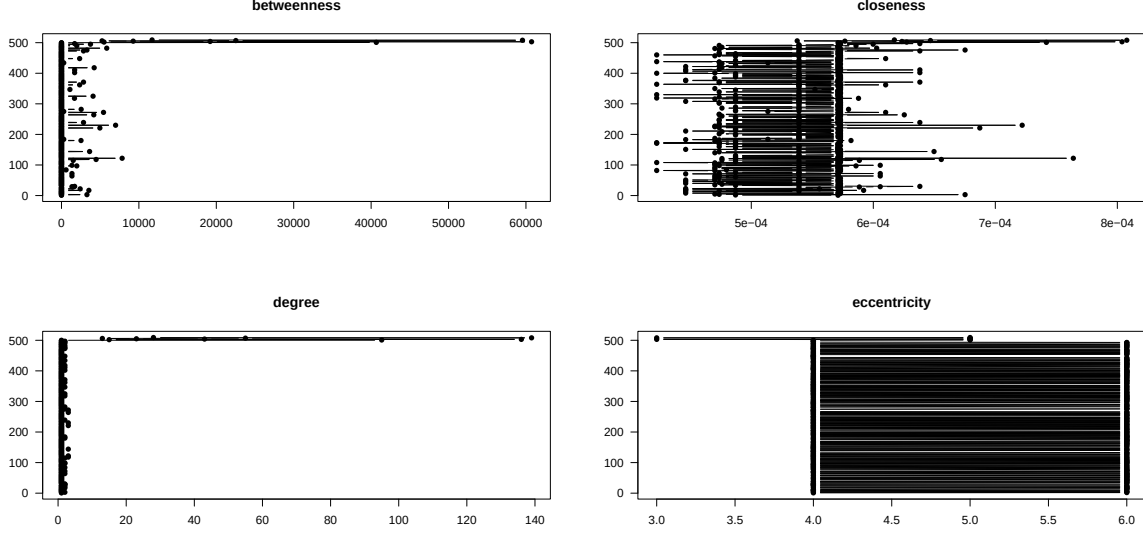
Figure 1: Network Visualisation



Note: Each colour represents a different partition of the network. The partitions' nodes are scaled to their relative degrees. Users with many different loan types are larger than those with single loan uses. Similarly, the size of second partition

nodes represents how many users belong to each.

Figure 2: Network Centrality Distribution Plots



3 Research Rationale

Our overall study analyses borrowers' behaviours, however, the studies are differentiated in their scope; the first study observes dyad-level relations among borrowers to understand how shared attributes affect borrowers' behaviours. This is inherently useful because behavioural finance aims to understand how individuals behave in *aggregate* (Ayal et al., 2018). Typically, the Quadratic Assignment Procedure is selected for dyad-level observations because basic linear regressions such as Ordinary Least Squares are fundamentally flawed and biased when using non-independent observations (Simpson, 2001).

Recognising this, studies such as Zuo (n.d.) assume that groups of borrowers have dependencies, and accordingly, use clustering methods such as K-Means or Fuzzy Clustering to understand behaviour. However, these methods are unsuitable when we require a distribution of estimates to make inferences from our results. Moreover, studying individuals as dyads rather than large clusters provides finer granularity in results. Therefore, QAP regression is the most appropriate choice for this study: it allows us to control for the non-independence among dyads, derive a statistical distribution of estimates through matrix permutation, and draw valid inferences about the relationship between borrower similarities and loan-use behaviour. This makes it both theoretically and methodologically well-suited to our research objectives and dataset.

The second study extends the analysis from a dyadic similarities to the structural formation of borrower-to-loan-use connections, which can identify the behavioural mechanisms that influence borrowers' specific loan use mixes. While QAP can account for exogenous influences, it cannot analyse higher-order network patterns such as clustering while accounting for simultaneous interactions between borrowers and loan uses. Using the Exponential Random Graph Model allows us to analyse how local behavioural tendencies aggregates to a global network structure.

Recent evidence highlights the need to focus on structural approaches in P2P lending. Liu, Baals, Osterrieder, & Hadji-Misheva (2024) studies borrowers’ centrality with the Bondora P2P dataset, modelling edges through borrower-to-borrower similarity metrics. Ultimately, they find that a borrower’s position is highly significant for their probability of default, using centrality as an exogenous variable in a logistic regression. In contrast, ERGM allows us to move away from predictive association and instead account for endogenous factors, allowing us to understand the likelihood that borrowers’ selection of loan uses are random. Consequently, we can better understand how behavioural phenomena such as familiarity and debt account aversion can translate into *systemic* patterns of credit use.

As we aim to understand how two distinct groups of nodes interact, a bipartite approach is highly suitable. Specifically, Stivala, Wang, & Lomi (2025) shows that in bipartite networks, terms such as cycle(4) can be effective in capturing clustering as in hypothesis 3, whereas it would fail in one-mode projected networks. Furthermore, the flexibility of ERGM also allows for modelling exogenous impacts such as heterophily in gender-based loan use mixes, as in hypothesis 6, and covariate age effects as in hypothesis 5.

4 Results

4.1 Study 1 - QAP Linear Regression

Figure Figure 3

Figure 3: QAP Coefficient Plots

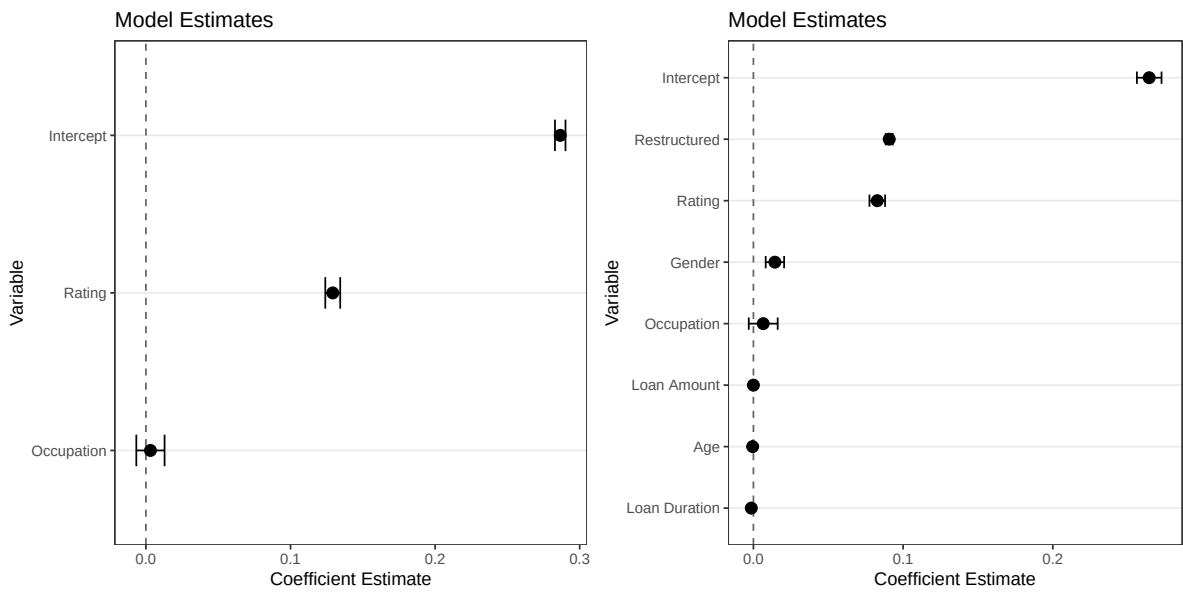


Table 5

Table 4

Table 5: QAP Modelling Results

	M1 Unstd.	M2 Std.	M3 Unstd.	M4 Std.
Intercept	0.286*** (0.002)	0.001 (0.002)	0.264*** (0.004)	0.001 (0.002)
Rating	0.129*** (0.003)	0.048*** (0.002)	0.083*** (0.003)	0.034*** (0.002)
Occupation	0.003 (0.005)	−0.001 (0.002)	0.006 (0.005)	−0.001 (0.002)
Loan Amount			−0.000*** (0.000)	−0.036*** (0.002)
Age			−0.001 (0.000)	−0.006 (0.002)
Gender			0.014 (0.003)	0.004 (0.002)
Loan Duration			−0.001** (0.000)	−0.021** (0.002)
Restructured			0.091*** (0.001)	0.092*** (0.002)
R-squared	0.010	0.002	0.035	0.013

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Table 6

Table 7: ERGM Modelling Results

	Base ERGM	b1degree(1)	cycle(4)	Age	Gender
edges	-1.978*** (0.046)	-1.354*** (0.123)	-0.940*** (0.279)	-0.862** (0.291)	-0.538 (0.462)
b1degree1		3.309*** (0.224)	3.267*** (0.250)	3.260*** (0.244)	3.271*** (0.247)
cycle4			-0.141 (0.087)	-0.141 (0.086)	-0.147 (0.086)
b1factor.b1_gender.male				-0.181 (0.207)	-0.168 (0.203)
b1cov.b1_age					-0.008 (0.009)
AIC	3325.828	2674.435	2676.439	2672.013	2674.852
BIC	3332.238	2687.255	2695.668	2697.652	2706.901
Log Likelihood	-1661.914	-1335.218	-1335.219	-1332.006	-1332.426

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

4.2 Study 2 - ERGM

Figure Figure 4

Figure 4: ERGM Coefficient Plots

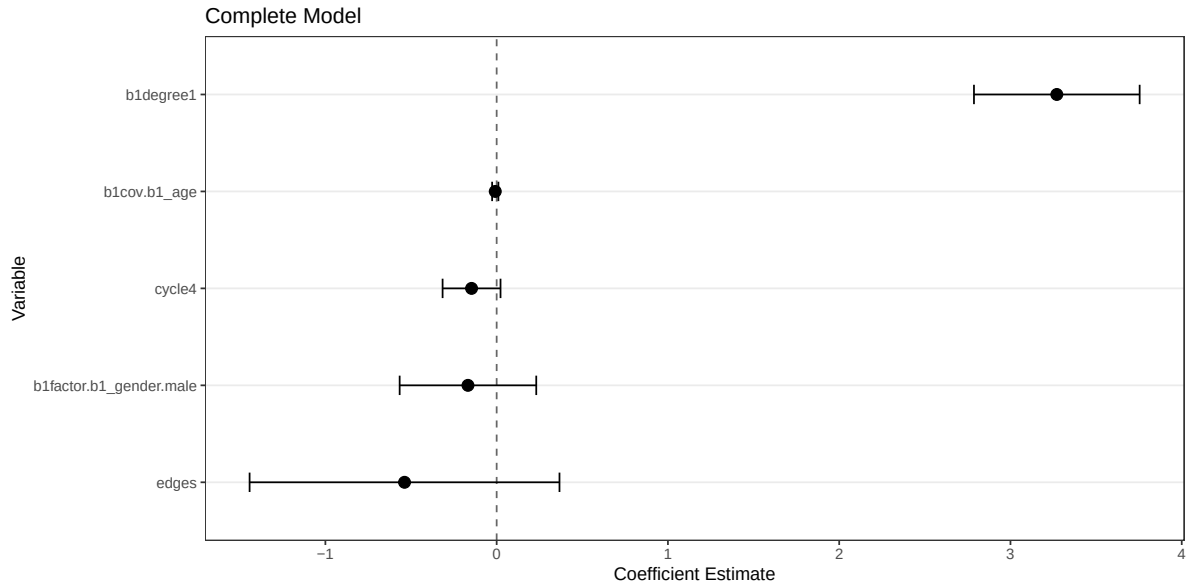


Table 7

A Supplements

A.1 Data Preprocessing Steps

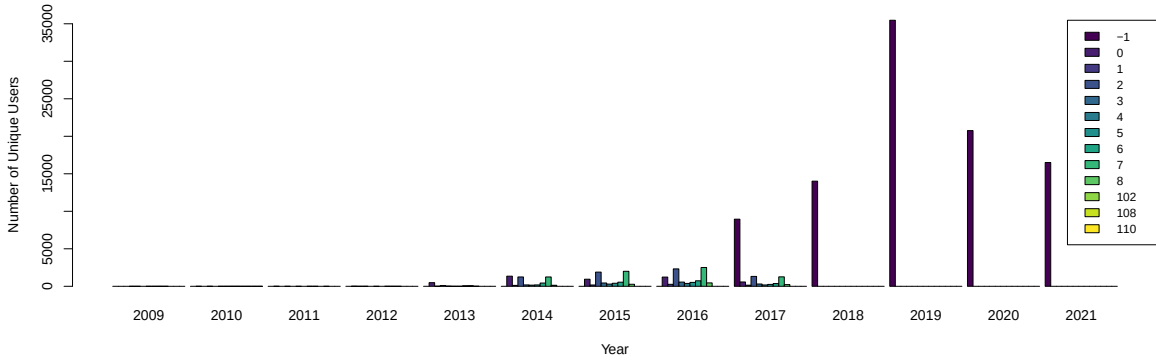
This section outlines the specific steps taken to process the data into the network object that was used in later analysis, outlining the rationale, where needed.

- Following the data import, only the following attributes were kept to make our processing steps more focused

```
keep_cols <- c("LoanId", "UserName", "Age", "Gender",  
              "Country", "Amount", "Interest", "LoanDuration",  
              "UseOfLoan", "Rating", "Restructured", "MonthlyPayment",  
              "OccupationArea", "BiddingStartedOn")
```

- We removed rows with any NA values for the selected attributes to ensure that we have a complete dataset
- Observing Figure 5, we see that between 2017-2021, there exists many -1 values for UseofLoan. This is because Bondora no longer provided this detail in later datasets. However, between 2014 and 2016 this data is available. Because of this, we restrict the time period to have the least noisy data. BiddingStartedOn was chosen to record the time, because the auction start date can be a good indication of activity around the loan.

Figure 5: Number of Unique Users per Loan Type per Year



- Then, we sampled 500 individuals out of the filtered dataset to ensure that our models run sufficiently well and converge. By choosing 500 individuals, we preserve importance structures within the network while adding sufficient statistical power.
- When adding labels to each loan use and occupation area, we keep any possible missing values because these do not indicate lower quality data. Instead, they can carry information about lenders distribute loans according to the quality of information presented by borrowers (Yao et al., 2019). Additionally, keeping missing labels is important to ensure that the edges are not superficially limited and that nodes can be isolates if they are indeed that way in reality.
- To create the network object for ERGM models:

- We first create an incidence matrix of size $n \times m$, borrower usernames by the loan use, where the values are the number of loans the user obtained for each loan use category. These values are then turned into binary because our modelling does not support the use of weighted edges.
- A bipartite network is created from this incidence matrix, where the first partition are unique users, and the second partition is the loan use associated among all loans of the user. Users can share more than more loan use type.
- We add age and gender attributes to each node of the first partition
- To create the relevant networks/matrices for QAP Regressions:
 - We first re-create the loan use incidence matrix, however, we keep the counts as the linear regression supports this.
 - * We then convert the loan use incidence matrix into an adjacency matrix of shape $n \times n$ through the transformation $\mathbf{X} \cdot \mathbf{X}^T$, which results in a matrix of shared loan uses weighted by the frequency of each user’s loan count within each loan use category.
 - * We then set the diagonal values of $\mathbf{X}$ to be zero to ensure that there are no loops within the network.
 - In accordance with this procedure, we create adjacency matrices for Credit Rating and Occupation Area which will be used as the main predictors in the QAP models.
 - * Credit Rating is a weighted adjacency matrix based on the loan count belonging to each user for each credit rating category
 - * Occupation Area is a binary adjacency matrix based on which occupation each user has reported to have held in the past when applying for each of their loans.
 - Additionally, we create a set of control variables to improve the validity and performance of the linear regression using Loan Amounts, Age, Gender, Loan Duration, and whether the loans have been restructured.
 - * Loan Amounts is a weighted adjacency matrix based on five bins across the range of possible loan amounts
 - * Age is an adjacency matrix based on differences in users’ ages
 - * Gender is a binary adjacency matrix based on users’ shared gender.
 - * Loan Duration is an adjacency matrix based on the differences in users’ loans average durations
 - * Restructured is a binary adjacency matrix based on whether the users share the fact that they have defaulted on any loans in the past. The users can either have shared both defaults and non-defaults, or both.

A.2 Distributions of Variables across Samples

Figure 6: Distribution of Loan Amount across Samples

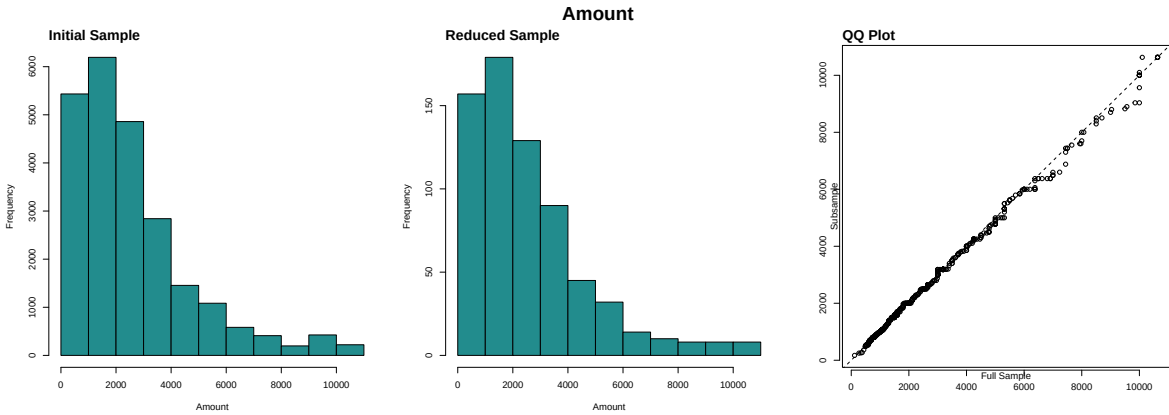


Figure 7: Distribution of Interest Rates across Samples

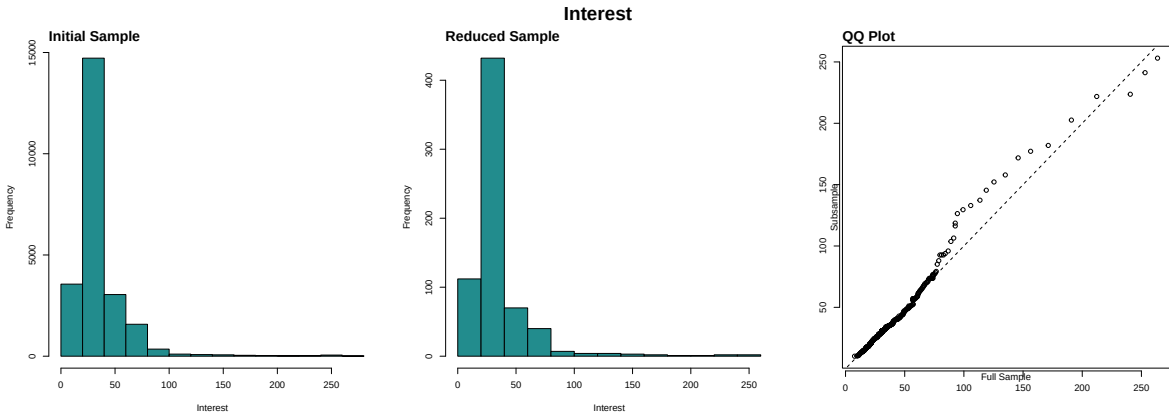


Figure 8: Distribution of Age across Samples

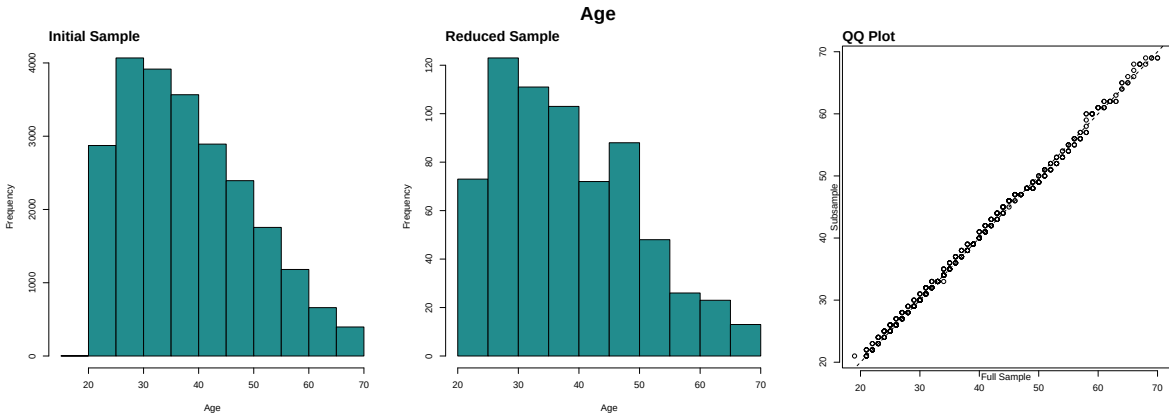


Figure 9: Distribution of Loan Duration across Samples

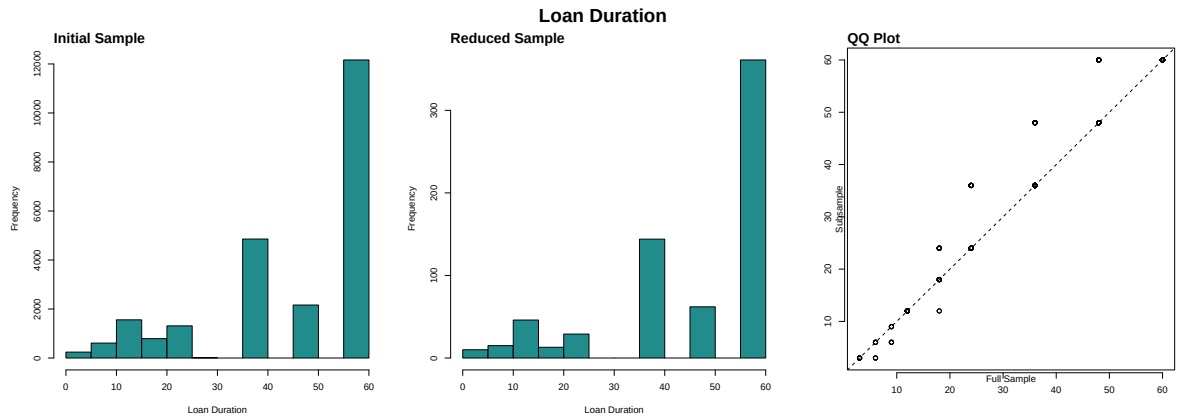


Figure 10: Distribution of Loan Purpose across Samples

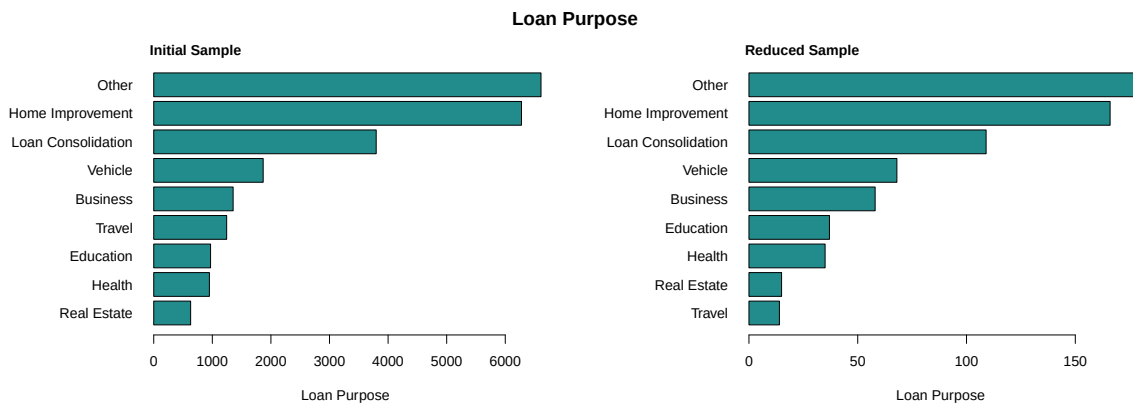


Figure 11: Distribution of Credit Ratings across Samples

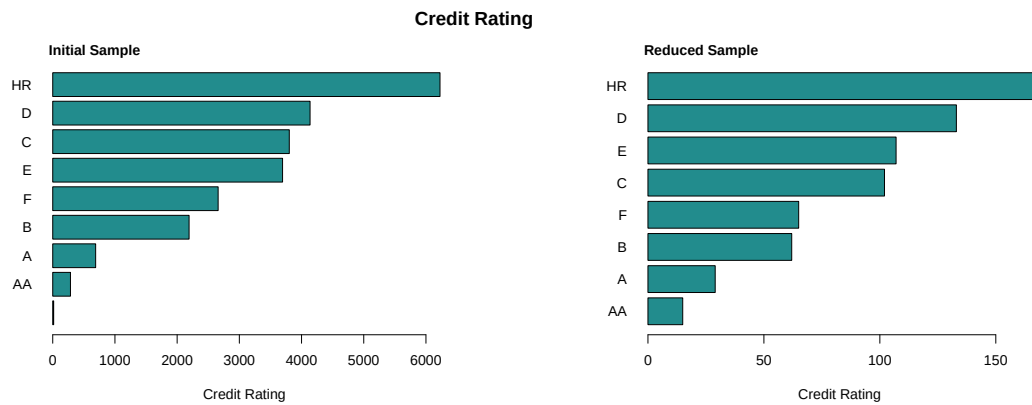
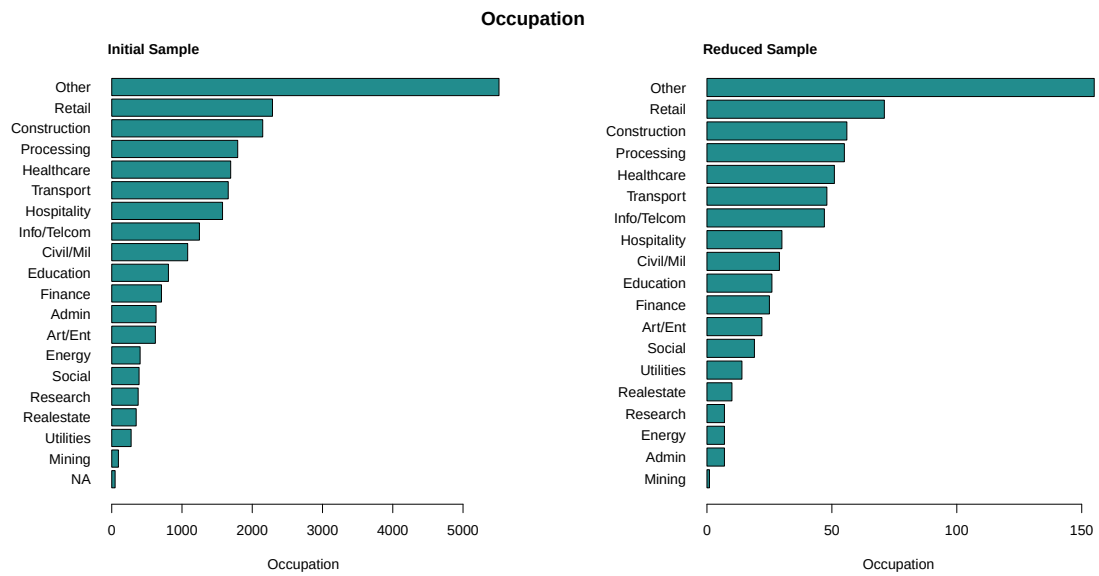


Figure 12: Distribution of Occupation Types across Samples



A.3 ERGM Results - Supplementary Tables

Table 8: ERGM Results in Probability Scale

Terms	Base	b1degree.1.	cycle.4.	Age	Gender
edges	0.122	0.205	0.281	0.297	0.369
b1degree1	-	0.965	0.963	0.963	0.963
cycle4	-	-	0.465	0.465	0.463
b1factor.b1_gender.male	-	-	-	0.455	0.458
b1cov.b1_age	-	-	-	-	0.498

A.4 Source Code - Data Preprocessing

```

1 # ----- #
2 install.packages("viridis") # For Colours
3 install.packages("here") # To locate files from RProj
4 set.seed(42)
5 # ----- #
6 # Import the Bondora P2P Dataset
7 obj_paths = "resources/objects/"
8 bondora_raw <- read.csv("dataset/LoanData_Bondora.csv")
9 raw_cols <- colnames(bondora_raw)
10 # ----- #
11 # Select Columns to Keep
12 keep_cols <- c("LoanId", "UserName", "Age", "Gender",
13               "Country", "Amount", "Interest", "LoanDuration",
14               "UseOfLoan", "Rating", "Restructured", "MonthlyPayment",
15               "OccupationArea", "BiddingStartedOn")
16
17 bondora <- bondora_raw[keep_cols]
18
19 # Change date format to the correct one
20 bondora$BiddingStartedOn <- as.POSIXct(bondora$BiddingStartedOn,
21                                       format = "%Y-%m-%d %H:%M:%S")
22 bondora$year <- as.numeric(format(bondora$BiddingStartedOn, "%Y"))
23
24 # Remove Rows with any NAs -> Complete Dataset Preferred
25 print(paste("NA Count |", sum(is.na(bondora)), "rows"))
26 bondora <- na.omit(bondora)
27 print(paste("NA Count |", sum(is.na(bondora)), "rows"))
28
29 # Observe the distribution of Loan Use over Time
30 unique_counts <- tapply(bondora$UserName,
31                         list(bondora$year, bondora$UseOfLoan),
32                         function(x) length(unique(x)))
33 unique_counts[is.na(unique_counts)] <- 0
34
35 barplot(t(unique_counts), # transpose so bars = loan types
36        beside = TRUE, # side-by-side bars
37        col = viridis::viridis(ncol(unique_counts)),
38        legend.text = colnames(unique_counts),
39        args.legend = list(x = "topright", cex = 0.8),
40        xlab = "Year",
41        ylab = "Number of Unique Users")
42 user_counts_plt <- recordPlot()
43 saveRDS(user_counts_plt,
44        here::here("resources", "objects", "preprocessing", "user_cnt_plt.Rds"))
45
46 # Restrict to Recent Time Period
47 bondora_test <- bondora[bondora$year > 2013 & bondora$year < 2017, ]
48
49 # See the distribution of LoanUses
50 see_counts <- function(var1, var2) {
51   counts <- table(var1, var2)
52   total_counts <- colSums(counts)
53   prop <- total_counts / sum(total_counts)
54
55   return(list(total_counts, round(prop, 2)))
56 }
57 before_counts <- see_counts(bondora_test$UserName, bondora_test$UseOfLoan)
58 barplot(before_counts[[1]])
59
60 # Number of unique users in the subsample
61 unique_users <- unique(bondora_test$UserName)
62 sample_users <- sample(unique_users, 500)
63 bondora_clean <- bondora_test[bondora_test$UserName %in% sample_users, ]
64
65 after_counts <- see_counts(bondora_clean$UserName, bondora_clean$UseOfLoan)

```

```

66 | barplot(after_counts[[1]])
67 |
68 | # DEPRECATED METHOD
69 | # Filter People with 2+ Loans
70 | #user_counts <- table(bondora_test$UserName)
71 | #multi_users <- names(user_counts[user_counts > 2])
72 | #bondora_clean <- bondora_test[bondora_test$UserName %in% multi_users, ]
73 |
74 | # DEPRECATED METHOD
75 | # Remove Users with only One Loan
76 | #user_counts <- table(bondora_clean$UserName)
77 | #multi_users <- names(user_counts[user_counts > 5])
78 | #bondora_clean <- bondora_clean[bondora_clean$UserName %in% multi_users, ]
79 | #bondora_test <- bondora_raw[bondora_raw$UserName %in% multi_users, ]
80 |
81 | # See if Ratings are Properly Encoded
82 | unique(bondora_clean$Rating)
83 | bondora_clean <- bondora_clean[bondora_clean$Rating != "", ]
84 |
85 | # Extract UseofLoan Types and Turn into Factor
86 | bondora_clean$UseOfLoan_factor <- as.factor(bondora_clean$UseOfLoan)
87 | unique(bondora_clean$UseOfLoan_factor)
88 |
89 | loan_use_labels <- c(
90 |   "-1" = "NA",
91 |   "0" = "Loan Consolidation",
92 |   "1" = "Real Estate",
93 |   "2" = "Home Improvement",
94 |   "3" = "Business",
95 |   "4" = "Education",
96 |   "5" = "Travel",
97 |   "6" = "Vehicle",
98 |   "7" = "Other",
99 |   "8" = "Health",
100 |   "110" = "Other Business",
101 |   "102" = "Undefined Business",
102 |   "108" = "Undefined Business"
103 | )
104 | # Change Labels for Cleaned Dataset
105 | bondora_clean$UseOfLoan_factor <- loan_use_labels[
106 |   as.character(bondora_clean$UseOfLoan)]
107 |
108 | # Change Labels for Uncleaned Dataset
109 | bondora_test$UseOfLoan_factor <- loan_use_labels[
110 |   as.character(bondora_test$UseOfLoan)]
111 |
112 | # Add labels to the OccupationArea Variable
113 | levels(as.factor(bondora_clean$OccupationArea)) # view codes
114 |
115 | occupation_labels <- c(
116 |   "-1" = "NA",
117 |   "1" = "Other",
118 |   "2" = "Mining",
119 |   "3" = "Processing",
120 |   "4" = "Energy",
121 |   "5" = "Utilities",
122 |   "6" = "Construction",
123 |   "7" = "Retail",
124 |   "8" = "Transport",
125 |   "9" = "Hospitality",
126 |   "10" = "Info/Telcom",
127 |   "11" = "Finance",
128 |   "12" = "Realestate",
129 |   "13" = "Research",
130 |   "14" = "Admin",
131 |   "15" = "Civil/Mil",

```

```

132 "16" = "Education",
133 "17" = "Healthcare",
134 "18" = "Social",
135 "19" = "Art/Ent",
136 "20" = "Agriculture",
137 "21" = "Forestry/Fish"
138 )
139 # store original
140 bondora_clean$occupation_code <- bondora_clean$OccupationArea
141 bondora_clean$occupation_label <- occupation_labels[
142   as.character(bondora_clean$OccupationArea)]
143
144 bondora_test$occupation_code <- bondora_test$OccupationArea
145 bondora_test$occupation_label <- occupation_labels[
146   as.character(bondora_test$OccupationArea)]
147 # ----- #
148 # Observe Descriptive Statistics
149
150 cols <- viridis::viridis(30)
151
152 # Make function to save plots
153 save_plot <- function(plt_nam) {
154   plt <- recordPlot()
155   saveRDS(plt, here::here("resources", "objects", "preprocessing",
156     paste0(plt_nam, ".Rds")))
157 }
158
159 # Make function to consistently plot comparisons
160 plot_desc_hists <- function(df1, df2, col_name, type) {
161
162   par(mfrow=c(1,3))
163
164   hist(df1[[col_name]], xlab=type, col=cols[15], main="", breaks=10)
165   mtext("Initial Sample", side=3, adj=0, line=0.25, cex=1, font=2)
166
167   hist(df2[[col_name]], xlab=type, col=cols[15], main="", breaks=10)
168   mtext("Reduced Sample", side=3, adj=0, line=0.25, cex=1, font=2)
169
170   qqplot(df1[[col_name]], df2[[col_name]], main="", cex=1,
171     xlab="Full Sample", ylab="Subsample", line=0.25)
172   abline(0, 1, lty=2)
173
174   mtext("QQ Plot", side=3, adj=0, line=0.25, cex=1, font=2)
175
176   mtext(type, outer = TRUE, line = -2, side=3, cex = 1.3, font = 2)
177
178   # Reset plot window
179   par(mfrow=c(1,1), mar=c(5,4,4,2)+0.1)
180 }
181
182 plot_desc_bar <- function(df1, df2, col_name, type) {
183
184   par(mfrow=c(1,2), mar=c(5,10,4,2))
185
186   barplot(sort(table(df1[[col_name]]), decreasing = F),
187     xlab=type, col=cols[15], horiz=TRUE, las=1)
188   mtext("Initial Sample", side=3, adj=0, line=0.25, cex=1, font=2)
189
190   barplot(sort(table(df2[[col_name]]), decreasing = F),
191     xlab=type, col=cols[15], horiz=TRUE, las=1)
192   mtext("Reduced Sample", side=3, adj=0, line=0.25, cex=1, font=2)
193
194   mtext(type, outer = TRUE, line = -2, side=3, cex = 1.3, font = 2)
195
196   # Reset plot window
197   par(mfrow=c(1,1), mar=c(5,4,4,2)+0.1)

```

```

198 }
199
200 plot_desc_hists(bondora_test, bondora_clean, "Amount", "Amount")
201 save_plot("hist_amt")
202
203 plot_desc_hists(bondora_test, bondora_clean, "Interest", "Interest")
204 save_plot("hist_int")
205
206 plot_desc_hists(bondora_test, bondora_clean, "LoanDuration", "Loan Duration")
207 save_plot("hist_loandur")
208
209 plot_desc_hists(bondora_test, bondora_clean, "MonthlyPayment", "Monthly Payment")
210 save_plot("hist_monpmt")
211
212 plot_desc_hists(bondora_test, bondora_clean, "Age", "Age")
213 save_plot("hist_age")
214
215 plot_desc_bar(bondora_test, bondora_clean, "UseOfLoan_factor", "Loan Purpose")
216 save_plot("bar_loanuse")
217
218 plot_desc_bar(bondora_test, bondora_clean, "Rating", "Credit Rating")
219 save_plot("bar_rating")
220
221 plot_desc_bar(bondora_test, bondora_clean, "occupation_label", "Occupation")
222 save_plot("bar_occupation")
223
224 # Get Tabular Summary Statistics
225 tab_comps <- function(df1, df2, cols) {
226   stats <- c("Mean"=mean, "Median"=median, "Std. Dev."=sd, "Min"=min, "Max"=max)
227
228   get_stats <- function(d) {
229     t(sapply(d[cols], function(x)
230       sapply(stats, function(f) round(f(x, na.rm=TRUE), 2))
231     ))
232   }
233
234   df1_stats <- get_stats(df1)
235   df2_stats <- get_stats(df2)
236
237   out <- rbind(
238     cbind(DataFrame = "Initial Sample",
239           Variable = rownames(df1_stats), df1_stats),
240     cbind(DataFrame = "Reduced Sample",
241           Variable = rownames(df2_stats), df2_stats)
242   )
243   rownames(out) <- NULL
244   as.data.frame(out)
245 }
246
247 tab_results <- tab_comps(bondora_test, bondora_clean,
248   c("Amount", "Interest", "Age", "LoanDuration"))
249
250 knitr::kable(tab_results)
251
252 # Save table for use in the Report
253 saveRDS(tab_results,
254   file=paste0(obj_paths, "preprocessing/", "summary_table.Rds"))
255 # ----- #
256 # Convert Dataset into Incidence Matrix to form Network Object (for ERGM)
257 bondora_slim <- bondora_clean
258
259 # Create the Incidence Matrix for Use of Loan
260 bondora_matrix <- table(
261   bondora_slim$UserName, bondora_slim$UseOfLoan)
262 bondora_matrix[bondora_matrix > 0] <- 1 # Given that ergm.counts fails with GOF
263

```

```

264 # Create network object with counts as Edge attribute
265 bondora_net <- network::network(
266   bondora_matrix, directed=FALSE, bipartite=nrow(bondora_matrix),
267   ignore.eval = FALSE, names.eval="frequency", loops=FALSE)
268
269 # Set the bipartite Attribute - UNNECESSARY GIVEN bipartite=length(n)
270 len <- dim(bondora_matrix)[1]
271 len_b2 <- dim(bondora_matrix)[2]
272 b_indicator <- c(rep(1,len),rep(2,len_b2))
273 network::set.vertex.attribute(bondora_net,
274                               "bipartite",value=b_indicator)
275
276 # Extract Partition 2 Labels
277 loan_use <- levels(bondora_clean$UseOfLoan_factor)
278
279 # Create Loan Type Attribute for Partition 2
280 b2_loantype <- rep(NA, len)
281 b2_loantype <- c(b2_loantype, loan_use)
282
283 if (length(b2_loantype) == network::network.size(bondora_net)) {
284   network::set.vertex.attribute(
285     bondora_net, "b2_loantype", value = b2_loantype)
286 }
287
288 # Add Age Vertex Attribute to B1
289 age <- bondora_clean$Age[match(
290   rownames(bondora_matrix), bondora_clean$UserName)]
291 b1_age <- c(age, rep(NA, len_b2))
292 network::set.vertex.attribute(
293   bondora_net, "b1_age", value=b1_age
294 )
295
296 # Add Gender Vertex Attribute to B1
297 gender <- bondora_clean$Gender[match(
298   rownames(bondora_matrix), bondora_clean$UserName)]
299 unique(gender) # Check to see if encoded properly
300 gender <- ifelse(gender == 0, "male", "female")
301 b1_gender <- c(gender, rep(NA, len_b2))
302 network::set.vertex.attribute(
303   bondora_net, "b1_gender", value = b1_gender
304 )
305
306 # Save the network and data frame object
307 saveRDS(bondora_slim, file=paste0(obj_paths, "preprocessing/", "bondora_df.Rds"))
308 saveRDS(bondora_net, file=paste0(obj_paths, "preprocessing/", "bondora_net.Rds"))
309 # ----- #
310 # First, create Incidence Matrix again but with Frequency Counts
311 loan_use_matrix <- table(
312   bondora_slim$UserName, bondora_slim$UseOfLoan)
313
314 # Get the Adjacency Matrix for Loan Use Similarity (Dependent QAP Variable)
315 adj_mat_loan_use <- loan_use_matrix %*% t(loan_use_matrix)
316 # Remove self weights to remove any loops
317 diag(adj_mat_loan_use) <- 0
318
319 # Get the Adjacency Matrix for Credit Rating Similarity (Predictor in QAP)
320 incidence_rating <- table(bondora_slim$UserName, bondora_slim$Rating)
321 adj_mat_rating <- incidence_rating %*% t(incidence_rating)
322 diag(adj_mat_rating) <- 0
323
324 # Get Adjacency Matrix for Occupation Similarity (Predictor in QAP, Binary)
325 incidence_occupation <- table(bondora_slim$UserName,
326                               bondora_slim$occupation_label)
327 adj_mat_occupation <- incidence_occupation %*% t(incidence_occupation)
328 adj_mat_occupation[adj_mat_occupation > 0] <- 1
329 diag(adj_mat_occupation) <- 0

```

```

330
331 # Get the Adjacency Matrix for Loan Amount (Control in QAP) - BINARY
332 # First, bin the Loan Amounts
333 bondora_slim$Amount_bins <- cut(
334   bondora_slim$Amount, breaks=c(0,2000,4000,6000,8000,10000),
335   labels = c(1:5)
336 )
337 incidence_amount_bins <- table(bondora_slim$UserName, bondora_slim$Amount_bins)
338 adj_mat_amount_bins <- incidence_amount_bins %>% t(incidence_amount_bins)
339 diag(adj_mat_amount_bins) <- 0
340
341 # Continous Absolute Difference Approach
342 avg_amount <- tapply(bondora_slim$Amount, bondora_slim$UserName, mean)
343 adj_mat_amount_diff <- outer(avg_amount, avg_amount,
344   FUN = function(x,y) abs(x - y))
345 diag(adj_mat_amount_diff) <- 0
346
347 # Get Matrix for Differences in Age
348 incidence_age <- table(bondora_slim$UserName, bondora_slim$Age)
349 borrower_ages <- as.numeric(colnames(incidence_age)[max.col(incidence_age)])
350 names(borrower_ages) <- rownames(incidence_age)
351 adj_mat_age <- outer(borrower_ages, borrower_ages,
352   FUN = function(x, y) abs(x - y))
353 rownames(adj_mat_age) <- colnames(adj_mat_age) <- names(borrower_ages)
354
355 # Get Adjacency Matrix for (same) Gender
356 incidence_gender <- table(bondora_slim$UserName, bondora_slim$Gender)
357 adj_mat_gender <- incidence_gender %>% t(incidence_gender)
358 # Make the matrix binary for homophily
359 adj_mat_gender <- ifelse(adj_mat_gender > 0, 1, 0)
360 diag(adj_mat_gender) <- 0
361
362 # Get Adjacency Matrix for Differences in Average Loan Duration
363 borrower_loandur <- sapply(tapply(bondora_slim$LoanDuration,
364   bondora_slim$UserName, unique),
365   mean)
366 adj_mat_loandur_diff <- outer(borrower_loandur, borrower_loandur,
367   FUN=function(x,y) abs(x-y))
368 diag(adj_mat_loandur_diff) <- 0
369
370 # Get Adjacency Matrix for Homophily in Restructure of Loans
371 incidence_restructure <- table(bondora_slim$UserName, bondora_slim$Restructured)
372 adj_mat_rest <- incidence_restructure %>% t(incidence_restructure)
373 diag(adj_mat_rest) <- 0
374
375 # Save the objects for the QAP Regression in different Script
376 qap_paths = paste0(obj_paths, "/qap/")
377 saveRDS(b_indicator, file=paste0(
378   "resources/objects/preprocessing/indicator.Rds"))
379
380 saveRDS(adj_mat_loan_use, file=paste0(qap_paths, "adj_mat_loanuse.Rds"))
381 saveRDS(adj_mat_occupation, file=paste0(qap_paths, "adj_mat_occup.Rds"))
382 saveRDS(adj_mat_rating, file=paste0(qap_paths, "adj_mat_rating.Rds"))
383 saveRDS(adj_mat_amount_diff, file=paste0(qap_paths, "adj_mat_amtdiffs.Rds"))
384 saveRDS(adj_mat_age, file=paste0(qap_paths, "adj_mat_agediffs.Rds"))
385 saveRDS(adj_mat_gender, file=paste0(qap_paths, "adj_mat_gender.Rds"))
386 saveRDS(adj_mat_loandur_diff, file=paste0(qap_paths, "adj_mat_loandurdiffs.Rds"))
387 saveRDS(adj_mat_rest, file=paste0(qap_paths, "adj_mat_rest.Rds"))
388 # ----- #

```

scripts/bondora_preprocessing.R

A.5 Source Code - QAP Linear Regression

```
1 # ----- #
2 # NOTE: NOT THE LATEST QAP ANALYSIS!
3
4 # If not already Installed
5 install.packages("viridis") # For Colours
6 install.packages("here") # To locate files from RProj
7 install.packages("gridExtra") # For multiple ggplot2 plots
8
9 # Set colour palette
10 cols <- viridis::viridis(30)
11
12 # Ensure repeatability
13 set.seed(42)
14 # ----- #
15 # Load Relevant Files
16 qap_path="resources/objects/qap/"
17
18 loan_use_mat <- readRDS(paste0(qap_path, "adj_mat_loanuse.RDS"))
19 rating_mat <- readRDS(paste0(qap_path, "adj_mat_rating.RDS"))
20 amt_diffs_mat <- readRDS(paste0(qap_path, "adj_mat_amtdiffs.RDS"))
21 age_diffs_mat <- readRDS(paste0(qap_path, "adj_mat_agediffs.RDS"))
22 gender_mat <- readRDS(paste0(qap_path, "adj_mat_gender.RDS"))
23 loandur_diffs_mat <- readRDS(paste0(qap_path, "adj_mat_loandurdiffs.RDS"))
24 rest_mat <- readRDS(paste0(qap_path, "adj_mat_rest.RDS"))
25 occup_mat <- readRDS(paste0(qap_path, "adj_mat_occup.Rds"))
26 # ----- #
27 # Create function to determine significance from t-value statistic
28 t_to_stars <- function(t) {
29   stars <- rep("", length(t))
30   stars[abs(t) >= 1.96] <- "*"
31   stars[abs(t) >= 2.576] <- "**"
32   stars[abs(t) >= 3.291] <- "***"
33
34   return(stars)
35 }
36
37 # Make function to save plots
38 save_qap_plot <- function(plt_nam) {
39   plt <- recordPlot()
40   saveRDS(plt, here::here("resources", "objects", "qap",
41                           paste0(plt_nam, ".Rds")))
42 }
43
44 # Make function to run QAPs automatically
45 run_QAP <- function(y_mat, xlist, varnames, model_name) {
46
47   # Run QAP Linear Regression
48   model <- sna::netlm(y = y_mat, x = xlist, nullhyp = "qapspp", reps = 3000)
49   model$names <- varnames
50   summary(model)
51
52   # Obtain results from Model and Name them
53   results <- model$coefficients
54   names(results) <- varnames
55
56   # Add significance stars to model results
57   results_sig <- paste(round(results, 3), t_to_stars(model$tstat))
58
59   # Save Model
60   saveRDS(model, here::here("resources", "objects", "qap",
61                             paste0(model_name, ".Rds")))
62
63   # Prepare function output
64   list_names <- c("model", "results", "results_sig")
65   items <- list(model, results, results_sig)
```



```

66 | names(items) <- list_names
67 |
68 | return(items)
69 | }
70 |
71 | # Make function to Plot QAP Coefficients
72 | plot_coef_qap <- function(model,
73 |                             model_name = "Model Estimates",
74 |                             x_axis_label = "Coefficient Estimate") {
75 |
76 |   # Extract Coefficients and SEs from ERGM
77 |   coefs <- model$coefficients
78 |   ses <- model$coefficients / model$tstat
79 |
80 |   # Make DF
81 |   df <- data.frame(
82 |     term = model$names,
83 |     estimate = coefs,
84 |     se = ses
85 |   )
86 |
87 |   # 95% Conf. Interval calculation (z-score 1.96)
88 |   df$lower <- df$estimate - 1.96 * df$se
89 |   df$upper <- df$estimate + 1.96 * df$se
90 |
91 |   # Significance indicator: CI does not cross zero
92 |   df$sig <- df$lower * df$upper > 0
93 |   df$sig <- as.character(df$sig) # Convert to character to make it work
94 |
95 |   # 3. Reorder terms by estimate (Base R factor reordering)
96 |   order_index <- order(df$estimate)
97 |   df$term <- factor(df$term, levels = df$term[order_index])
98 |
99 |   # 4. Create the ggplot (Uses ggplot2:: explicitly, as requested)
100 |   ggplot2::ggplot(df, ggplot2::aes(x = estimate, y = term)) +
101 |     ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color = "grey40") +
102 |     ggplot2::geom_errorbarh(ggplot2::aes(xmin = lower, xmax = upper),
103 |                           height = 0.2) +
104 |     ggplot2::geom_point(size = 3) +
105 |     ggplot2::theme_bw() +
106 |     ggplot2::theme(
107 |       panel.grid.major.x = ggplot2::element_blank(),
108 |       panel.grid.minor.x = ggplot2::element_blank()
109 |     ) +
110 |     ggplot2::labs(
111 |       x = x_axis_label,
112 |       y = "Variable",
113 |       title = model_name,
114 |       #color = "Significance"
115 |     )
116 | }
117 | # Make Function to Plot Diagnostics
118 | # DEPRECATED SINCE PROGRESS MEETING 3
119 | qap_diags <- function(model, title, filename) {
120 |   resid <- model$model$residuals
121 |   fitted <- model$model$fitted.values
122 |
123 |   # Set viewing window to two plots
124 |   par(mfrow=c(1,2))
125 |
126 |   # Residuals Plot
127 |   hist(resid, xlab="Residuals",
128 |        main=title)
129 |
130 |   # Fitted vs Residuals
131 |   plot(fitted, resid, xlab="Fitted Values", ylab="Residuals",
132 |        main=title)

```

```

132
133 # Capture and Save the Plot
134 save_qap_plot(filename)
135
136 # Reset plot view
137 par(mfrow=c(1,1))
138 }
139
140 var_names <- c("Intercept", "Rating", "Occupation", "Loan Amount", "Age",
141               "Gender", "Loan Duration", "Restructured")
142 main_pred_names <- c("Intercept", "Rating", "Occupation")
143 main_pred_vars <- list(rating_mat, occup_mat)
144 all_pred_vars <- list(rating_mat, occup_mat, amt_diffs_mat, age_diffs_mat,
145                      gender_mat, loandur_diffs_mat, rest_mat)
146 # ----- #
147 # Basic QAP Linear Regression 1 - Unstandardised + No Controls
148 qap_m1 <- run_QAP(loan_use_mat, main_pred_vars, main_pred_names, "qap_m1")
149 summary(qap_m1$model)
150
151 # Plot the QAP Model Coefficients and Save it
152 qap_m1_coef_plot <- plot_coef_qap(qap_m1$model)
153 saveRDS(qap_m1_coef_plot,
154         here::here("resources", "objects", "qap", "qap_m1_coef_plot.Rds"))
155
156 # Run Diagnostics Plots - DEPRECATED SINCE PROGRESS MEETING 3 SUGGESTIONS
157 #qap_diags(qap_m1, "QAP LR - Unstandardised + No Controls", "qap_m1_diags")
158
159 # ----- #
160 # Basic QAP Linear Regression 2 - Unstandardised + Controls
161 qap_m3 <- run_QAP(loan_use_mat, all_pred_vars, var_names, "qap_m3")
162
163 summary(qap_m3$model)
164
165 # Plot the QAP Model Coefficients and Save it
166 qap_m3_coef_plot <- plot_coef_qap(qap_m3$model)
167 saveRDS(qap_m3_coef_plot,
168         here::here("resources", "objects", "qap", "qap_m3_coef_plot.Rds"))
169
170 # Arrange the Plots together
171 qap_coef_plots <- gridExtra::grid.arrange(qap_m1_coef_plot, qap_m3_coef_plot,
172                                           ncol = 2)
173
174 # Save the Plots
175 saveRDS(qap_coef_plots,
176         here::here("resources", "objects", "qap", "qap_coef_plots.Rds"))
177
178 # Run Diagnostics Plots - DEPRECATED SINCE PROGRESS MEETING 3 SUGGESTIONS
179 #qap_diags(qap_m3, "QAP LR - Unstandardised + Controls", "qap_m3_diags")
180
181 # ----- #
182 # Collect Results in Table
183
184 # Extract R2 from the netlm object from the Console output
185 # It was too difficult to calculate R2 from scratch and netlm does not
186 # provide this as an output directly from the netlm object
187 calc_r2 <- function(model) {
188   out <- capture.output(model)
189   r2_line <- grep("Multiple R-squared", out, value = TRUE)
190   r2 <- as.numeric(
191     sub(".*Multiple R-squared:\\\\s*([0-9\\.]+).*", "\\1", r2_line))
192   return(r2)
193 }
194
195
196 # Make function to get details of netlm manually
197 extract_netlm <- function(model) {

```

```

198   tr <- texreg::createTexreg(
199     coef.names = model$names,
200     coef = model$coefficients,
201     se = model$coefficients / model$tstat,
202     pvalues = model$pgreqabs,
203     gof.names = c("R-squared"),
204     gof = (calc_r2(model)),
205     gof.decimal = c(TRUE)
206   )
207   return(tr)
208 }
209
210 # View the Table of Results
211 titles <- c("M1 | No Controls", "M2 | Controls")
212 extracted_models <- lapply(list(qap_m1, qap_m3), extract.netlm)
213 texreg::screenreg(extracted_models, custom.model.names = titles, digits = 3)
214
215 # Save the table for use in the Report
216 saveRDS(list(extracted_models, titles),
217         here::here("resources", "objects", "qap", "qap_tables.Rds"))
218 # ----- #

```

scripts/qap_network_analysis.R

A.6 Source Code - ERGM Network Analysis

```
1 # ----- #
2 # NOTE: NOT THE LATEST ERGM ANALYSIS! FOR REFERENCE ONLY
3
4 # If not already Installed
5 install.packages("viridis")      # For Colours
6 install.packages("Rglpk")        # Additional solver for ERGMs
7 install.packages("here")         # To locate files from RProj
8
9 # Import the Network and Other Object
10 ergm_path <- "resources/objects/ergm/"
11 bondora_net <- readRDS(here::here(
12   "resources", "objects", "preprocessing", "bondora_net.Rds"))
13 b_indicator <- readRDS(here::here(
14   "resources", "objects", "preprocessing", "indicator.Rds"))
15
16 # Set colour palette
17 cols <- viridis::viridis(30)
18
19 # Determine acceptable core count
20 n_cores <- parallel::detectCores() - 3 # Leave some out for other processes
21 print(paste("You have", n_cores, "usable cores"))
22
23 # Repeatability
24 set.seed(42)
25
26 # Save plots
27 save_ergm_plot <- function(plt_nam) {
28   plt <- recordPlot()
29   saveRDS(plt, here::here("resources", "objects", "ergm",
30     paste0(plt_nam, ".Rds")))
31 }
32 # ----- #
33 # Copy network for plotting
34 bondora_plot <- bondora_net
35
36 # Get node type for plotting
37 type_indicator <- ifelse(b_indicator == 2, TRUE, FALSE)
38 shape <- ifelse(type_indicator, "square", "circle")
39 network::set.vertex.attribute(bondora_plot, "shape", shape)
40
41 # Get Category Count for Vertex Size
42 counts <- sna::degree(bondora_plot)
43 counts_att <- ifelse(type_indicator, log(counts)*4, counts*2.5)
44 network::set.vertex.attribute(bondora_plot, "size", counts_att)
45
46 # Colours for the Node Types
47 plot_cols <- ifelse(type_indicator, cols[5], cols[30])
48 network::set.vertex.attribute(bondora_plot, "color", plot_cols)
49
50 # Legend Plotting
51 type_legend <- ifelse(type_indicator, "Borrowers", "Loan Type")
52 type_legend <- as.factor(type_legend)
53
54 # Plot the Network
55 plot(snafun::to_igraph(bondora_plot),
56   #main = "Bipartite User-LoanUse",
57   edge.arrow.size = 0.3,
58   edge.color = rgb(0,0,0, alpha = 0.35),
59   vertex.frame.color = "black",
60   vertex.label = NA,
61   vertex.frame.size = 3,
62   edge.curved = FALSE,
63   layout=igraph::layout.fruchterman.reingold)
64 legend("bottomleft",
65   legend = levels(type_legend),
```

```

66     inset = c(0.15, 0.01),
67     col = c(cols[30], cols[5]),
68     pch = c(16, 15),
69     title = "Node Partitions",
70     title.font = 2,
71     cex = 1,                # Increase the text size
72     pt.cex = 2,            # Increase the point symbol size
73     box.lwd = 1,           # Thin box border
74     box.col = "black",     # Box color
75     bty = "o"              # Use a box around legend
76 )
77 save_ergm_plot("network_plot")
78
79 # Summary Statistics and Save Them
80 density <- snafun::g_density(bondora_net)[1]
81 centralization <- snafun::g_centralize(bondora_net)[1]
82 vertices <- snafun::count_vertices(bondora_net)[1]
83 edges <- snafun::count_edges(bondora_net)[1]
84 dist <- snafun::g_mean_distance(bondora_net)[1]
85
86 net_names <- c("Vertex Count", "Edge Count", "Density", "Centralization",
87               "Mean Distance")
88 net_stats <- c(vertices, edges, density, centralization, dist)
89 net_stats <- sapply(net_stats, function(x) round(as.numeric(x), 2))
90
91 net_summary <- data.frame(Statistic = net_names,
92                           "Measure" = net_stats)
93 knitr::kable(net_summary)
94 saveRDS(net_summary, here::here("resources", "objects", "ergm", "net_summary.Rds"))
95
96 # Plot Network Summary Statistics
97 snafun::plot_centralities(bondora_net)
98 save_ergm_plot("network_plots")
99 # ----- #
100 # Make function to calculate probabilities from log odds
101 lodds_to_prob <- function(l_odd) {
102   return(exp(l_odd) / (1 + exp(l_odd)))
103 }
104 # Make function to save ERGM object
105 save_ergm <- function(object, id) {
106   saveRDS(object, file=here::here(
107     "resources", "objects", "ergm", paste0(id, ".Rds")))
108 }
109 # Make function to conduct ERGMs automatically
110 auto_ergm <- function(model, mcmc, name) {
111
112   # Conducts the GOF Diagnostics and then saves the model,
113   # mcmc diagnostics and gof object in a list.
114   # This list can be imported as an .RDS object into the R environment
115
116   # Diagnostics
117   if (mcmc) {
118     ergm::mcmc.diagnostics(model)
119   }
120
121   # The GOF must be adjusted otherwise it takes too long
122   # We do not limit the GOF by changing its range of parameters
123   gof <- ergm::gof(model,
124                     control = ergm::control.gof.ergm(
125                       nsim = 200,
126                       MCMC.burnin = 5000,
127                       MCMC.interval = 1000,
128                       parallel = n_cores,
129                       parallel.type = "PSOCK"
130                     ))
131

```

```

132 # Return List to view each item separately
133 result <- list(model, gof)
134 names(result) <- c("model", "gof")
135 save_ergm(result, paste0(name, "_panel"))
136
137 return(result)
138 }
139 # Make Function to do Coefficient Plot
140 plot_coef <- function(model,
141                       model_name = "Model Estimates",
142                       x_axis_label = "Coefficient Estimate") {
143
144   # Extract Coefficients and SEs from ERGM
145   coefs <- coef(model)
146   ses <- sqrt(diag(vcov(model)))
147
148   # Make DF
149   df <- data.frame(
150     term = names(coefs),
151     estimate = coefs,
152     se = ses
153   )
154
155   # 95% Conf. Interval calculation (z-score 1.96)
156   df$lower <- df$estimate - 1.96 * df$se
157   df$upper <- df$estimate + 1.96 * df$se
158
159   # Significance indicator: CI does not cross zero
160   df$sig <- df$lower * df$upper > 0
161   df$sig <- as.character(df$sig) # Convert to character to make it work
162
163   # 3. Reorder terms by estimate (Base R factor reordering)
164   order_index <- order(df$estimate)
165   df$term <- factor(df$term, levels = df$term[order_index])
166
167   # 4. Create the ggplot (Uses ggplot2:: explicitly, as requested)
168   ggplot2::ggplot(df, ggplot2::aes(x = estimate, y = term)) +
169     ggplot2::geom_vline(xintercept = 0, linetype = "dashed", color = "grey40") +
170     ggplot2::geom_errorbarh(ggplot2::aes(xmin = lower, xmax = upper),
171                           height = 0.2) +
172     ggplot2::geom_point(size = 3) +
173     ggplot2::theme_bw() +
174     ggplot2::theme(
175       panel.grid.major.x = ggplot2::element_blank(),
176       panel.grid.minor.x = ggplot2::element_blank()) +
177     ggplot2::labs(
178       x = x_axis_label,
179       y = "Variable",
180       title = model_name,
181       #color = "Significance"
182     )
183 }
184 # ----- #
185 # Find max degree
186 (max_deg <- max(summary(bondora_net ~ b2factor("b2_loantype"))))
187 # ----- #
188 # Base Model + GOF
189 formula_base_model <- bondora_net ~ edges
190 base_ergm <- ergm::ergm(formula_base_model)
191 base_ergm_panel <- auto_ergm(base_ergm, mcmc = FALSE, name = "ergm_base")
192 snafun::stat_plot_gof(base_ergm_panel$gof)
193 models = list(base_ergm)
194 texreg::screenreg(models)
195 # ----- #
196 # Iteration 1 + MCMC Diagnostics + GOF
197 model_1_params <- bondora_net ~ edges +

```

```

198 # See if individuals tend to pick one loan type
199 bldegree(1)
200
201 model_1 <- ergm::ergm(
202   model_1_params,
203   #constraints = ~ bd(minout = 0, maxout = 80),
204
205   control = ergm::control.ergm(
206     # Greater burn-in for cleaner result
207     MCMC.burnin = 20000,
208     # Greater sample size for greater stability
209     MCMC.samplesize = 100000,
210     seed = 42,
211     MCMC.interval = 1000,
212     # Only needed for convergence pvals to improve
213     MCMLE.maxit = 45,
214     # Smaller steps for stability
215     MCMLE.steplength = 0.25,
216     parallel = n_cores,
217     parallel.type = "PSOCK"
218   )
219 )
220
221 model_1_panel <- auto_ergm(model=model_1, mcmc=TRUE, name="ergm_m1")
222 model_1_panel$gof
223 snafun::stat_plot_gof(model_1_panel$gof)
224 texreg::screenreg(list(base_ergm, model_1))
225
226 # Plot Coefficients
227 plot_coef(ergm_m1$model, model_name = "Baseline + bldegree")
228 # ----- #
229 # Iteration 2 + MCMC Diagnostics + GOF
230 model_2_params <- bondora_net ~ edges + bldegree(1) +
231   # See if there is clustering around pairs of loan types
232   cycle(4)
233
234 model_2 <- ergm::ergm(
235   model_2_params,
236   #constraints = ~ bd(minout = 0, maxout = 80),
237
238   control = ergm::control.ergm(
239     # Greater burn-in for cleaner result
240     MCMC.burnin = 20000,
241     # Greater sample size for greater stability
242     MCMC.samplesize = 100000,
243     seed = 42,
244     MCMC.interval = 1000,
245     # Only needed for convergence pvals to improve
246     MCMLE.maxit = 45,
247     # Smaller steps for stability
248     MCMLE.steplength = 0.25,
249     parallel = n_cores,
250     parallel.type = "PSOCK"
251   )
252 )
253
254 model_2_panel <- auto_ergm(model=model_2, mcmc=TRUE, name="ergm_m2")
255 snafun::stat_plot_gof(model_2_panel$gof)
256 model_2_panel$gof
257 models <- list(base_ergm, model_1, model_2)
258 texreg::screenreg(models)
259
260 # Plot Coefficients
261 plot_coef(model_2_panel$model, model_name = "Baseline + bldegree + cycle(4)")
262 # ----- #
263 # Iteration 3 + MCMC Diagnostics + GOF

```

```

264 model_3_params <- bondora_net ~ edges + b1degree(1) + cycle(4) +
265
266 # See if there is gender effect
267 blfactor("b1_gender")
268
269 model_3 <- ergm::ergm(
270   model_3_params,
271   #constraints = ~ bd(minout = 0, maxout = 80),
272
273   control = ergm::control.ergm(
274     # Greater burn-in for cleaner result
275     # Cycles mixed with the rest of the terms requires more burnin samples
276     MCMC.burnin = 20000,
277     # Greater sample size for greater stability
278     MCMC.samplesize = 100000,
279     seed = 42,
280     # Reducing interval to reduce complexity between intervals
281     MCMC.interval = 1000,
282     # Only needed for convergence pvals to improve
283     MCMLE.maxit = 25,
284     # Smaller steps for stability
285     MCMLE.steplength = 0.25,
286     parallel = n_cores,
287     parallel.type = "PSOCK"
288   )
289 )
290
291 model_3_panel <- auto_ergm(model=model_3, mcmc=TRUE, name="ergm_m3")
292 snafun::stat_plot_gof(model_3_panel$gof)
293 model_3_panel$gof
294 models <- list(base_ergm, model_1, model_2, model_3)
295 texreg::screenreg(models)
296
297 # Plot Coefficients
298 plot_coef(model_3_panel$model,
299           model_name = "Baseline + b1degree + cycle(4) + blfactor(gender)")
300 # ----- #
301 # Iteration 4 + MCMC Diagnostics + GOF
302 model_4_params <- bondora_net ~ edges + b1degree(1) + cycle(4) +
303   blfactor("b1_gender") +
304
305 # See if higher ages make a difference
306 blcov("b1_age")
307
308 model_4 <- ergm::ergm(
309   model_4_params,
310   #constraints = ~ bd(minout = 0, maxout = 80),
311
312   control = ergm::control.ergm(
313     # Greater burn-in for cleaner result
314     MCMC.burnin = 20000,
315     # Greater sample size for greater stability
316     MCMC.samplesize = 100000,
317     seed = 42,
318     MCMC.interval = 1000,
319     # Only needed for convergence pvals to improve
320     MCMLE.maxit = 45,
321     # Smaller steps for stability
322     MCMLE.steplength = 0.25,
323     parallel = n_cores,
324     parallel.type = "PSOCK"
325   )
326 )
327
328 model_4_panel <- auto_ergm(model=model_4, mcmc=TRUE, name="ergm_m4")
329 model_4_panel$gof

```



```

330 snafun::stat_plot_gof(model_4_panel$gof)
331 models <- list(base_ergm, model_1, model_2, model_3, model_4)
332 texreg::screenreg(models)
333
334 # Plot Coefficients
335 m4_ergm_coef_plot <- plot_coef(model_4_panel$model,
336                               model_name = "Complete Model")
337 # Save Plot
338 saveRDS(m4_ergm_coef_plot,
339         here::here("resources", "objects", "ergm", "m4_coef_plot.Rds"))
340 ggplot2::ggsave("m4_coef_plot.png",
341                plot = m4_ergm_coef_plot,
342                device = "png",
343                path = here::here("resources", "objects", "ergm"))
344 # ----- #
345 # Collect Models in Table 1: Raw results
346 headers <- c("Base ERGM", "b1degree(1)", "cycle(4)", "Age", "Gender")
347 texreg::knitreg(models, custom.model.names = headers,
348                 digits = 3)
349
350 # Save the Table to use in Report
351 saveRDS(list(models, headers),
352         here::here("resources", "objects", "ergm", "ergm_table.Rds"))
353
354 # Padding
355 pad <- function(x, max) {
356   return(c(unname(x), rep("-", max-length(x))))
357 }
358
359 # Collect Models in Table 2: Log Odds Results
360 max_ergms <- length(model_4$coefficients)
361 df_prob <- data.frame(
362   Terms = names(model_4$coefficients),
363   "Base" = pad(round(lodds_to_prob(base_ergm$coefficients), 3), max_ergms),
364   "b1degree(1)" = pad(round(lodds_to_prob(model_1$coefficients), 3), max_ergms),
365   "cycle(4)" = pad(round(lodds_to_prob(model_2$coefficients), 3), max_ergms),
366   "Age" = pad(round(lodds_to_prob(model_3$coefficients), 3), max_ergms),
367   "Gender" = pad(round(lodds_to_prob(model_4$coefficients), 3), max_ergms)
368 )
369 knitr::kable(df_prob)
370
371 # Save Table to use in Report
372 saveRDS(df_prob,
373         here::here("resources", "objects", "ergm", "ergm_prob_tables.Rds"))
374 # ----- #

```

scripts/ergm_network_analysis.R

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B Technology Statement

During the preparation of this work, we used ChatGPT in order to generate select parts of the R script utilised to process the dataset. Specifically, the tool was used to transform the processed dataset into a format that igraph would accept as a network object. Additionally, some R functions written by the group were improved with ChatGPT's suggestions. No AI tool was utilised to independently write parts of the report. The following parts of the assignment were affected/generated by AI tool usage: **INTRODUCTION**; The tool was utilised to evaluate the validity of certain theoretical concepts and refine them. **DATASET**; the data described within this section was processed partly by some code drafted by ChatGPT and edited by the group. After using this tool/service, **Group 7** evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, **Group 7** takes full responsibility for the content of their work.